

Muhammad Mehdi

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LinkedIn Profile

OBJECTIVE

I aim to apply my skills in GIS, Remote Sensing, GeoAI and satellite hydrology to understand how climate change is reshaping water resources in the Hindu Kush–Karakoram–Himalaya (HKH) region. My goal is to pursue graduate research at a leading research university that combines satellite gravimetry (GRACE/GRACE-FO), multi-source Earth observation and GeoAI for Earth observation: measuring changes in water storage, glaciers and snow, using deep learning and geospatial AI to extract reliable information from large satellite datasets, and testing how well global datasets describe the water cycle in mountain basins where ground measurements are scarce. Ultimately, I aspire to become a geospatial scientist who bridges advanced geospatial technologies with practical solutions for water security and climate resilience.

RESEARCH INTERESTS

- Satellite gravimetry (GRACE/GRACE-FO) and terrestrial water storage change.
- Water balance of glacier- and snow-fed mountain basins; glacier and snow mass change in High Mountain Asia.
- Evaluation of global precipitation, evapotranspiration and runoff datasets; GeoAI and multi-source Earth observation for environmental monitoring.

EDUCATION & TRAINING

September 2023 - June 2027 (expected) | Lahore, Pakistan
Bachelor of Science in Geography, University of the Punjab, Lahore
Grade: CGPA 3.57 out of 4.00 (up to 6th semester) | **University website**

RESEARCH EXPERIENCE

2026 - Current | Supervisor: Assist. Prof. Dr. Adeel Ahmad, University of the Punjab
BS Thesis: Testing Global Hydroclimate Products Against Satellite Gravimetry: A Water-Balance Benchmark for the Glacier-Fed Upper Indus Basin (In progress; manuscript in preparation)

- Testing whether global rainfall, evaporation and runoff datasets add up to the water-storage change measured by the GRACE/GRACE-FO satellites (JPL, CSR and GSFC solutions) in the Upper Indus Basin above Tarbela (about 172,700 km², 2002–2025).
- Comparing precipitation (ERA5-Land, IMERG, MSWEP, CHIRPS) and evapotranspiration (GLEAM, ERA5-Land, GLDAS, TerraClimate, PML-V2) products, while accounting for datasets that share the same inputs, water held in Tarbela reservoir, and glacier areas (RGI 7.0).
- Done so far: delineated the basin from HydroBASINS, cross-checked it with MERIT Hydro and removed the closed Pangong Tso basin; checked every monthly dataset for gaps, units and errors, which uncovered and corrected a documented error in ERA5-Land data for September 2022 to February 2024. The analysis in Python is now under way.

April 2026 - Current | Lahore, Pakistan
Undergraduate Research Intern, GeoScape Analytics Lab (GSAL)

- Working on snow cover and glacier-fed runoff in the Indus basin using Google Earth Engine, Python and machine learning.
- Applied remote sensing and spatial analysis to Urban Dry and Wet Island effects and groundwater dynamics.
- Applied GeoAI and machine-learning methods (CNNs, TensorFlow) to satellite datasets for climate-change monitoring.
- Co-developed GRIPS, the lab's literature-search app, and built web maps, dashboards and the lab's official website.

Lab website: **View website**

MANUSCRIPTS (SUBMITTED AND IN PREPARATION)

Data-Driven Approaches for GRACE/GRACE-FO Downscaling: A Systematic Review (2021–2026)
Sarwar, N., Waseem, H., Ahmad, A., Bukhari, S. S. B., Komal, I., **Mehdi, M.**, Tipu, H. M. B., Shafi, S., Asim, S., Javed, M. A., & Waqar, M. Systematic review. **(Submitted to a Springer Nature journal, October 2026)**

Climate and Glacier Dynamics Fragment the Karakoram Mass-Balance Anomaly
Hussain, I., …, **Mehdi, M.**, et al. Research article. **(Submitted to the Journal of Hydrology)**

Spatially Adaptive Weighting for Composite Vulnerability Indexes in Heterogeneous Agricultural Landscapes of Punjab, Pakistan
Ahmad, A., **Mehdi, M.**, & Maqsood, U. In: *Spatial Intelligence: Designing Composite Indexes through Geocomputation*. Taylor & Francis Group. Book chapter. **(Submitted)**

GIS Applications for Monitoring and Pattern Identification
Gilani, H., Bukhari, S. S. B., Tipu, H. M. B., **Mehdi, M.**, & Ahmad, A. In: *Elements of Smog Monitoring and Mitigation*. Elsevier. Book chapter. **(Submitted)**

GEOAI, WEB & APPLICATION PROJECTS

GRIPS: Geospatial Research Index of Pakistan Studies (Co-developer, GeoScape Analytics Lab, 2026)
Free literature-search web app for Pakistan-focused GIS, remote sensing and environmental research. It searches several scholarly databases (OpenAlex, Semantic Scholar, CrossRef, arXiv), removes duplicates and exports results to CSV and BibTeX. Version 1.0.0 archived on Zenodo. **View app**

CarbonScope-TX, FortyGuard Hackathon'26
AI application for climate resilience analysis built on FortyGuard's hyperlocal temperature API. **View project**

HumanScape (Demography Atlas)
Interactive global atlas of population, births and deaths (1950–2100) from UN World Population Prospects. **View atlas**

University of the Punjab, Interactive Campus Map
First interactive web map of the Quaid-e-Azam Campus, Lahore. **View live map**

TECHNICAL SKILLS

Satellite Hydrology Data: GRACE/GRACE-FO mascons (JPL, CSR, GSFC) | ERA5-Land | IMERG | MSWEP | CHIRPS | GLEAM | GLDAS | TerraClimate | PML-V2

GIS & Remote Sensing: Google Earth Engine | QGIS | ArcGIS Pro | HydroBASINS | MERIT Hydro | RGI 7.0

Python (geospatial): geemap | leafmap | xarray | GeoPandas | Rasterio | rioxarray | GDAL | Shapely | NumPy | pandas | Matplotlib (vector, raster and NetCDF/HDF5 gridded data) | Machine Learning

Other Programming & Data: R | JavaScript (Google Earth Engine) | NetCDF/HDF5 data handling | Data quality checking

Web & Documentation: Interactive web maps and dashboards | LaTeX & Overleaf

CERTIFICATIONS

Scanned copies of all certificates listed below are available here: **View certificate folder**

- Sep 2026 | **Introduction to GeoAI**, YouthMappers Academy
- Aug 2026 | **FortyGuard Hackathon'26: Certificate of Demonstrated Excellence**, FortyGuard (awarded for CarbonScope-TX)
- Jul 2026 | **CNNs with TensorFlow: Basics of Machine Learning**, Coursera. **Verify certificate**
- Jun 2026 | **Introduction to Python**, Coursera. **Verify certificate**
- May 2025 | **Make an Impact with Modern Geo Apps**, Esri
- Mar 2025 | **Chinese for Beginners**, Peking University (Coursera). **Verify certificate**
- Dec 2024 | **Getting Started with CyberGIS**, University of Illinois Urbana-Champaign (Coursera). **Verify certificate**

CONFERENCES, WORKSHOP & FIELD PROJECT

April 2026 | Online, ESGRS & Mohammed Bin Rashid Space Centre (MBRSC)
Open-Source GeoAI for Earth Observation (Webinar), led by Dr. Qiusheng Wu

December 2025 | Lahore
Public Opinion on Climate Change in Lahore, Field Survey
Surveyed 40 respondents and produced a technical report with recommendations.

October 2025 | Institute of Geography, University of the Punjab
One-Day Workshop on Differential GPS (DGPS)

December 2024 | Institute of Geography, University of the Punjab
1st National Geographical Conference: "Sustainable Earth Resources and Planning"

HONOURS AND AWARDS

October 2025 | Institute of Geography, University of the Punjab
2nd Position, World Habitat Day Competition ("Sustainable Cities and Urban Geography")

LANGUAGE SKILLS

Urdu: Native | **English:** Professional working proficiency

REFERENCE

Dr. Adeel Ahmad
In-charge, GeoScape Analytics Lab (GSAL); Assistant Professor, Institute of Geography, University of the Punjab, Lahore
Email: adeel.geog@pu.edu.pk | **Relationship:** Research Supervisor